

Alan Huang

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EDUCATION

Stanford University

Bachelor of Science in Computer Science

- GPA: 3.96 / 4.00

Master of Science in Computer Science (Coterminal)

September 2024 – May 2028

Stanford, CA

EXPERIENCE

Software Engineer Intern

Walmart Global Tech

- Engineered and deployed a Python automation service for the Walmart Fulfillment Services (WFS) team that consolidated queries from six backend fulfillment systems, enabling 40 seller-support agents to investigate order status and shipping exceptions 67% faster (15→5 minutes)
- Delivered 8 production bug fixes and authored 15 regression test cases for a Java order-tracking service processing over 500K daily orders, improving system reliability through peer-reviewed pull requests
- Collaborated with senior engineers and product managers to prioritize seller pain points, translating operational feedback into 7 roadmap features adopted by the fulfillment dashboard team

June 2026 – August 2026

Bentonville, AR

AI Research Intern

National Science Foundation (NSF)

- Selected for the NSF REU Site: Enhancing Undergraduate Experiences in Data and Mobile Cloud Security, focusing on cybersecurity, mobile/cloud computing, and AI-driven data security hosted at Purdue University
- Improved cross-dataset deepfake detection accuracy from 59.7% to 65.5% on the CelebDF benchmark by designing a diffusion-based Synthetic Domain Alignment technique, validated across FaceForensics++ and CelebDF
- Published as co-first author: *Robust Deepfake Detection via Synthetic Domain Alignment*, 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025)

June 2025 – August 2025

Indianapolis, IN

PROJECTS

OS: Crash-Consistent Filesystem & Encrypted Virtual Memory | C, C++

2026

- Implemented a Unix V6 filesystem supporting crash consistency through write-ahead logging and journal replay, guaranteeing metadata recovery after simulated mid-write failures
- Built a memory-mapped encrypted file system in C++ using the Clock page-replacement algorithm to manage 128 physical memory pages, achieving a 27% reduction in page faults compared to a naive FIFO baseline

Multi-Label Hate Crime Bias Classification | Python, scikit-learn, XGBoost

2026

- Designed an end-to-end machine learning pipeline for multi-label bias classification including preprocessing, imbalance mitigation, hyperparameter tuning, and model evaluation across five supervised algorithms
- Trained and benchmarked five classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost, majority baseline) to predict bias motivation category across 32,508 California hate crime incidents
- Addressed severe class imbalance (three rarest classes <3% of data) using SMOTE and stratified splitting, more than doubling macro F1 from 12.4% (majority-class baseline) to 25.0% with L2-regularized Logistic Regression

PUBLICATIONS

Alan Huang, Ian Yee, Shu Hu. *Robust Deepfake Detection via Synthetic Domain Alignment*. The 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025).

[DOI](#)

Alan Huang, Shaheer Raza, Yuni Xia. *Predicting Diabetes with Medical and Demographic Data*. The 6th International Conference on Computer Science, Engineering, and Education (CSEE 2025).

[DOI](#)

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript

Frameworks: FastAPI, Express, Node.js

Machine Learning: TensorFlow, XGBoost, scikit-learn, pandas, NumPy

Systems & Tools: Git, Linux, Supabase, Docker, Bash, PostgreSQL